Topological Sort

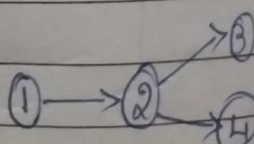
```
#include <stdio.h>

void topologicalSort (int v, int adj[v][v])
{
    int indegree [v], queue[v];
    int front =0, rear = -1;

    for (int i=0 ; i<v; i++)
        indegree [i]=0
    for (int i =0 ; i<v ; i++ )
    {
        for (int j=0; j<v; j++)
        {
            if (adj[i][j] ==1)
                indegree [j] ++ ;
        }
    }

    for (int i=0; i<v; i++ )
        if (indegree[i]==0)
            queue [++rear] = i;

    printf ("\n Topological order ");

    while (front <=rear)
    {
        int node = queue [front++];
        printf ("%d " node);
        for (int i=0 ; i<v; i++)
        {
            if (adj [node][i] ==1)
```

```
            {
                indegree[i]--;
                if (indegree[i]==0)
                {
                    queue[++rear] = i;
                }
            }
        }
    }
}

int main()
{
    int v, adj[10][10];

    printf("Enter the no of vertices: ");
    scanf("%d", &v);

    printf("\n Enter the adjacency matrix: ");

    for (int i=0; i<v; i++)
        for (int j=0; j<v; j++)
            scanf("%d", &adj[i][j]);

    topologicalSort(v, adj);
    return 0;
}
```

Output : Enter the no. of vertex: 4
Enter adjacency matrix :

0 1 0 0
0 0 1 1
0 0 0 0
0 0 0 0

Topological order
1 2 3 4



	1	2	3	4
1	0	1	0	0
2	0	0	1	1
3	0	0	0	0
4	0	0	0	0